Adjective Ordering Preferences: An Incremental RSA Account with Bayesian Inference

LMBayes Meeting 2026 — Discussion Handout

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Introduction

Speakers of diverse languages exhibit robust ordering preferences of adjectives (e.g., preferring "the big brown bag" over "the brown big bag"). A way to approach this regularity is through communicative efficiency: helping interlocutors coordinate quickly and reliably. Yet "efficiency" can be formalized from two perspectives that make different ordering predictions: a hierarchical, comprehension-oriented view in which modification is composed outward from the noun, and a linear, production-oriented view in which speakers benefit from placing highly informative material early in the unfolding word stream. The present project pursues a third, unified perspective: adjective ordering explained by *incremental pragmatic reasoning*, where a speaker produces one adjective at a time and the in-context informativeness of each adjective interacts with its inherent semantic precision.

The project has three interlocking components:

1. **Model.** An incremental variant of the Rational Speech Act (RSA) framework, implemented in NumPyro / JAX, with Bayesian inference over model parameters.
2. **Experiments.** Two complementary data sources—a slider-rating study measuring graded truth values, and a free-production experiment measuring ordering preferences.
3. **Simulation.** A broad parameter sweep exploring the behaviour of the model across context-size distributions, perceptual blur, and semantic-value parameters.

Discussion: Starting questions

- What is the minimal pragmatic mechanism that *derives* ordering preferences rather than stipulating them?
- What role does *inherent* vs. *contextual* informativeness play for different adjective classes?
- How do we best connect a Bayesian model to categorical production data?

1 Model

1.1 Representational primitives

An **object** is a triple $o = (s, c, f)$ where $s \in [1, 30]$ is size, $c \in \{0, 1\}$ colour match, and $f \in \{0, 1\}$ form match. A **context** $\mathcal{C}$ is a set of n_{obj} objects; the referent is always the largest object with $c = f = 1$.

Utterance space. We allow single-adjective and multi-adjective utterances built from three basic categories: D (dimension/size), C (colour), F (form). Combining lengths and orderings yields 15 surface forms: {D, DC, DCF, DF, DFC, C, CD, CDF, CF, CFD, F, FD, FDC, FC, FCD}.

1.2 Atomic semantics

Size (D). We adopt a *slack semantics* based on the Approximate Number System. A context-dependent threshold is computed as

$$\theta_k(\mathcal{C}) = x^{\text{mid}}_{\text{max}}(\mathcal{C}) - k\left(x^{\text{mid}}_{\text{max}}(\mathcal{C}) - x^{\text{mid}}_{\text{min}}(\mathcal{C})\right), \tag{1}$$

where $x^{\text{mid}}_{\text{min}}$ and $x^{\text{mid}}_{\text{max}}$ are the q_{low}- and q_{high}-quantiles of the contextual size distribution ($q_{\text{low}} = 0.2$, $q_{\text{high}} = 0.8$). The semantic value of applying D to object o is then

$$\mathrm{D}(o, \mathcal{C}) = \Phi\left(\frac{s_o - \theta_k(\mathcal{C})}{w_f\sqrt{s_o^2 + \theta_k(\mathcal{C})^2}}\right), \tag{2}$$

where Φ is the standard normal CDF and $w_f \geq 0$ is a *perceptual blur* parameter.

Colour (C) and Form (F). These features have determinate denotations, but we parameterise their semantic strength by a single *color/form semantic value* $\nu \in (0, 1)$ applied via a logistic link function.

1.3 Compositional meaning

Meanings compose *left to right* (reflecting incremental production): starting from a uniform prior $p_0(o)$, each successive adjective updates the listener's posterior:

$$p_{t+1}(o) \propto p_t(o) \cdot a_{t+1}(o, \mathcal{C}). \tag{3}$$

The *literal listener* $L_0(o \mid u, \mathcal{C})$ is the posterior after processing all adjectives in utterance u.

1.4 Utterance prior (cost)

We incorporate three cost signals into the utterance prior:

(i) **LM-derived cost:** $\text{cost}_{\text{LM}}(u) = -\log P_{\text{LM}}(u)^{\beta}$, where P_{LM} is a GPT-based prior over the 15 utterances and β controls sensitivity. Shorter, high-frequency forms (*e.g.* D) are cheaper.

(ii) **Subjectivity bias:** an additive penalty favouring subjective-first orderings (D-initial), captured by a bias parameter.

(iii) **Length cost:** utterance length enters implicitly via P_{LM}.

1.5 Speaker models

Global speaker $S_{\mathbf{G}}$. A standard RSA pragmatic speaker that evaluates a complete utterance's utility:

$$P_{S_{\text{G}}}(u \mid o, \mathcal{C}) \propto \exp\big[\alpha \cdot \log L_0(o \mid u, \mathcal{C}) - \lambda \cdot \text{cost}(u)\big]. \tag{4}$$

Incremental speaker S_{I}. Produces one adjective at a time; at step t it chooses the next adjective a_t given the prefix already produced. The updated-listener posterior after each step feeds into the next step's cost-weighted speaker distribution:

$$P_{S_{\mathrm{I}}}(a_1 \mid o, \mathcal{C}) \propto \exp\big[\alpha \cdot \log L_0(o \mid a_1, \mathcal{C}) - \lambda \cdot \mathrm{cost}(a_1)\big], \tag{5}$$

$$P_{S_{\mathrm{I}}}(a_t \mid a_{1:t-1}, o, \mathcal{C}) \propto \exp\big[\alpha \cdot \log p_t(o) - \lambda \cdot \mathrm{cost}(a_t \mid a_{1:t-1})\big]. \tag{6}$$

1.6 Two model variants & parameterisation

Variant	Description	Free parameters
Flat / global	Global speaker, uniform or LM prior; single set of parameters across all items	$\alpha, k, w_f, \nu, \beta$
Hierarchical / incremental	Incremental speaker; item-level random effects on k and ν; hyper-priors over means and variances	$\mu_k, \sigma_k, \mu_\nu, \sigma_\nu, \alpha, \beta, w_f$

Discussion: Model discussion

- Should w_f (perceptual blur) be inferred or fixed? Slider data suggest a sharp vs. blurred manipulation, but the mapping to a single scalar is unclear.
- The LM prior collapses utterance-order preferences and length. Should these be decoupled?
- Is the left-to-right compositionality assumption linguistically motivated or merely computationally convenient?

2 Experiments: Overall Design

Both experiments share the same factorial design:

Factor	Levels
F_1: **Relevant property**	Size-relevant (`zrdc`), Colour-relevant (`erdc`), Both-relevant (`brdc`)
F_2: **Sharpness**	Sharp (size is diagnostically large), Blurred (size is gradient/perceptually noisy)
Fixed: Referent = largest, colour-matching, form-matching object in a 6-object context	

Each item encodes a context as a 6×3 matrix (s, c, f) with the referent at index 0. Conditions outside the {`zrdc, erdc, brdc`} set (non-size-adjective manipulations) are excluded from modelling.

Figure 1: Illustration of an experiment item (placeholder). Left panel: a sharp-size context; right panel: a blurred-size context. The referent (circled) is the largest, colour-matching, form-matching object.

2.1 Slider-rating study (Experiment 1)

Task. Participants rated the *felicity* of each adjective description on a continuous slider (0–100) for a given context-referent pair. Ratings were collected for single-adjective (D, C, F) and two-adjective (DC, DF, CD,) descriptions.

Dependent variable. Normalised slider ratings, interpreted as empirical truth values $[\![u]\!]_{\text{emp}}(o, \mathcal{C}) \in [0, 1]$.

Key findings.

- Size ratings are markedly lower in the blurred condition than the sharp condition, validating the w_f manipulation.
- Colour and form ratings are high and stable across conditions.
- Ordering interactions: in size-relevant contexts, DC is rated higher than CD; the pattern reverses in colour-relevant contexts.

Modelling goal. Fit the incremental RSA literal listener to graded slider ratings in order to estimate k, w_f, and ν. These values serve as informative priors for the production model.

Key Result: Slider modelling

Posterior estimates: $k \approx$ **[TODO: X]**, $w_f \approx$ **[TODO: X]** (sharp), $w_f \approx$ **[TODO: X]** (blurred), $\nu \approx$ **[TODO: X]**. **[TODO: Insert MCMC summary / forest plot here.]**

Discussion: Slider-study discussion

- The slider task confounds *semantic acceptability* with *pragmatic felicity*. How should we interpret the ratings?
- Should we run a replication study with a larger sample? (A replication dataset is available; results not yet integrated.)
- Should k vary by condition or item, or is a single population-level value enough?

2.2 Free-production study (Experiment 2)

Task. Participants described the referent object to a hypothetical listener in as few words as possible, using only the provided adjectives. Responses were manually annotated into the 15 utterance categories.

Dependent variable. Categorical annotation $y_i \in \{1, \ldots, 15\}$ for each trial i.

Conditions. **[TODO: N participants, M items, L lists; exact numbers to be confirmed.]**

Key findings.

1. **Relevant-property effect.** Size-first orderings (D, DC, DCF) dominate in size-relevant contexts; colour-first (C, CD) dominate in colour-relevant contexts.
2. **Sharpness effect.** Size adjectives appear more frequently (including over-informatively) in the blurred condition than the sharp condition.
3. **Single adjective dominance.** D alone is the modal response across most conditions; multi-word utterances are more common when both properties are relevant.

Modelling challenges.

- *Categorical likelihood instability.* Some rare categories have near-zero model probability, driving the log-likelihood to $-\infty$. Solution: aggregate trials to empirical *soft distributions* per condition, and use a Dirichlet likelihood.
- *Sparse data per item.* Items with few observations show unreliable empirical distributions. Solution: hierarchical model with item-level random effects.
- *NUTS + discrete latents.* NUTS requires differentiability; all discrete variables (utterance type) must be marginalised.

Key Result: Production modelling

[TODO: Insert posterior forest plot / predictive check here.] Global model: $\alpha \approx$ **[TODO: X]**, $\beta \approx$ **[TODO: X]**. Incremental model: $\mu_k \approx$ **[TODO: X]**, $\mu_{w_f} \approx$ **[TODO: X]**.

Discussion: Production-study discussion

- Is the Dirichlet likelihood the right choice, or should we use a Multinomial with soft targets?
- How do we evaluate model fit beyond visual predictive checks? LOO-CV? WAIC?
- The production data includes a replication study. Should the two datasets be modelled jointly or separately?
- Should the LM prior (β) be inferred from the data, or is it better fixed from corpus statistics?

3 Simulation

3.1 Goals

The simulation has three purposes:

(1) Establish that the incremental speaker *generates* ordering preferences, not just accommodates them, across a broad range of contexts.

(2) Map out which parameter region produces a qualitative *big-before-colour* preference and compare to a *colour-before-big* region.

(3) Assess robustness: does the ordering preference persist under varying context size (n_{obj}), size-distribution spread (σ_{spread}), and perceptual blur (w_f)?

3.2 Design

Parameter	**Values swept**	**# levels**
Speaker model	Incremental, Global	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
ν (colour semantic value)	0.90, 0.92, 0.94, 0.96, 0.98	5
k (threshold)	0.10, 0.30, 0.50, 0.70, 0.90	5
w_f (perceptual blur)	0.1, 0.2, 0.3, 0.5, 0.8	5
Random context samples per combination		1000
Total rows		**6,000,000**

Each simulated trial computes communicative success $p(o_{\text{ref}} \mid u)$ for two key orderings (big blue > blue big) and records which ordering the speaker model assigns higher probability.

3.3 Key results

[TODO: Insert simulation plots here – e.g., proportion of big-first responses as a function of k, w_f, and n_{obj}.]

Key Result: Simulation findings

- The incremental speaker robustly prefers D-first over the majority of the parameter space when size is contextually informative.
- The threshold parameter k has the largest impact on ordering preferences: lower k (higher threshold) reduces size-first preference.
- The effect of n_{obj} is mediated by σ_{spread}: in narrow distributions, larger context size amplifies differences.
- The global speaker shows a qualitatively similar but weaker preference, suggesting the incremental mechanism is a contributing, not sole, factor.

Discussion: Simulation discussion

- The simulation uses random contexts; should we run it on the actual experimental contexts to bridge to the empirical data?
- What is the right summary statistic: mean communicative success, proportion of trials preferring one ordering, or something else?
- Does the simulation yield clear predictions about the replication data?

4 General Discussion

4.1 Summary

We have presented an incremental RSA account of adjective ordering preferences that unifies informativeness-based and subjectivity-based explanations. The model is fitted to two complementary datasets (slider ratings and free production) using NUTS-based Bayesian inference in NumPyro / JAX, and its qualitative behaviour is validated across a large-scale parameter simulation.

Key claims:

- A simple incremental production rule—choose the next adjective proportional to its *residual* informativeness—generates ordering preferences without a fixed hierarchy.
- Perceptual blur (w_f) modulates size-adjective usage, explaining the sharpness effect.
- LM-derived costs provide a grounded utterance prior that is consistent with observed single-adjective dominance.

4.2 Open issues for the write-up

1. **Model comparison.** A formal comparison of global vs. incremental speaker (LOO-CV or Bayes factors) is still outstanding.
2. **Replication dataset.** The slider-rating replication study needs to be integrated; modelling approach not yet finalised.
3. **Predictions vs. fits.** Are the parameter values recovered from the slider data good priors for the production model? A formal prior-predictive check for the production model is needed.
4. **LM prior validation.** The GPT-derived utterance prior should be evaluated against human acceptability judgments.
5. **Scope of adjective classes.** The model currently handles D, C, F; extension to other adjective classes (e.g., material, age) is not addressed.
6. **Individual differences.** The hierarchical model captures item-level variability but not participant-level variability.
7. **Write-up structure.** A tentative section order: Introduction → Background (RSA, adjective ordering) → Model → Experiments → Results → Simulation → General Discussion.

Discussion: For the meeting

- What is the right framing for the paper: pragmatics, psycholinguistics, or computational linguistics?
- Target venue? **[TODO: CogSci / Cognition / PLOS ONE / Glossa / ?]**
- What level of detail should the model section have for the target audience?
- Figures wish-list: parameter recovery plot, posterior predictive vs. empirical distributions per condition, simulation heatmap.

References

[TODO: Add references (BibTeX file TBD). Key citations to include: Frank & Goodman (2012), Bergen et al. (2016), Scontras et al. (2017), Brehm & Lam (2019), Piantadosi et al. (2012), Pyykkonen et al. (ANS semantics).]